# ECE 5746 Fall 2026
# Lab #3 – Automatic Place & Route (APR)

***Due 10/21/2026 (Wednesday) by 11:00PM, Submission to Canvas***

***15% penalty up to 2 hours late; 30% penalty up to 6 hours late;***
***50% penalty up to 24 hours late; all submissions later than 24 hours will receive 0 points.***

In Lab 2, you synthesized and explored the design space of the 16-bit multiplier.

As a continuum, in this Lab 3, you will explore the APR and Post-APR process with EDA tools.

Retrieve the synthesized netlist that was synthesized at 1.20 × $\text{clk_period}_{\text{min}}$ from Lab 2. You will use this netlist to go through APR in this lab.

**Automatic Place and Route (APR) Process**

- Before starting automatic place and route (APR) using Cadence Innovus tool, please copy all the required files mentioned in "Tutorial on Cadence Innovus.pdf" (at Canvas) to your run directory.
- Follow the sample Innovus APR script provided at: /classes/ece5746/projects/TutorialAPR/example_innovus.tcl
- Make necessary changes for your design and run the script to perform APR.
- Ensure to understand each command and corresponding options (e.g. run "man <command_name>" at the Innovus prompt, or refer to /opt/cadence/DDI231/INNOVUS231/doc/innovusTCR/innovusTCR.pdf file at the ecelinux server).
- Change relevant commands in the script to place your input ports on the left (west) side of the floorplan and the output ports on the right (east) side.
- Further documentation can be found at (from the ecelinux server):
    - Innovus user guide: /opt/cadence/DDI231/INNOVUS231/doc/innovusUG/innovusUG.pdf
    - Innovus text command reference: /opt/cadence/DDI231/INNOVUS231/doc/innovusTCR/innovusTCR.pdf
- Modify clock_period variable in var.tcl to use the same clock period as your synthesized design. Start APR with the same clock period of "1.20 × $\text{clk_period}_{\text{min}}$", and let's denote this as "**clk_per_syn**". Check if the post-route timing doesn't have any setup time violations with this same clock period. If yes, move on the next stage below. If not (this scenario is more likely), increase the clock_period value in the var.tcl file for APR until setup time violation is all clear (and let's denote the minimum APR clock period that doesn't have any setup time violation as "**clk_per_apr**"). In the report file, clearly report both "**clk_per_syn**" and "**clk_per_apr**".

    - This is just an exercise, and you will not be graded by the absolute values of "**clk_per_syn**" and "**clk_per_apr**". You can sweep several values, but there is no need for a rigorous sweep of the clock period.
- After your design is complete, you need to generate reports for area (from summary report) and timing. You can refer to the Innovus tutorial and additional documentation above, for the required commands.
    - Obtain the information of 'core area' and 'chip area' from summary report.
    - From the generated timing reports, you need to report the 3 worst paths for setup and hold time after post-route stage. You can find the worst path from your timing report and report it as **critical path delay** in your datasheet. (You wouldn't be graded by the value of the critical path delay, we are just asking you to report the delay by running the timing report.). Observe the setup timing report in path *timingReports/<top_level>_postRoute_all.tarpt.gz* and hold timing report in *timingReports/<top_level>_postRoute_all_hold.tarpt.gz*

**Post APR Process**

- Refer to "Tutorial on Post-APR" document at Canvas. Carefully follow the steps in the document. Be sure to edit your project specific design name and paths in *streamIn.sh* . Source the streamIn.sh file in ~/tsmcN16_ADFP directory where you will run your virtuoso.
- Import the gds file from Innvous into Cadence Virtuoso and perform do a visual check on layout.
- Run post-layout Verilog simulation with sdf back annotation (refer to the Tutorial on Post-APR", using the same testbench that you used in Lab 2.
- Make sure that you use the <top_level>.apr.v for post-layout Verilog simulation.

**Submission Guidelines**

**Strictly adhere to the guidelines. Stick to the given directory names and maintain hierarchy mentioned below.**

Put all files in one Linux directory with the following organization:

lab3_<net_id>

Documents/

Datasheet.pdf *// This should contain a clear table of # of standard cells, 'core area', 'chip area', 'clk_per_syn', 'clk_per_apr'*

*(Note: Complexity of your design is generally measured by number of cells, macros, frequency, and area)*

DC/

mult.<clk_period>.RVT.syn.v *// synthesized netlist that you used as APR input*

mult.<clk_period>.syn.sdc

Innovus/

mult.apr.v *// placed and routed netlist (Innovus output)*

mult.apr.physical.v *// placed and routed netlist with pg pins (Innovus output)*

mult.sdf *// sdf file (Innovus output)*

mult.final.tar *// tarred final design*

*// the compressed tar file should include the following files and directories (.enc files, corresponding .dat directory, .lef file, .lib file, .sdc file)*

*// make sure the design can be opened by placing them all in a sperate directory and running Innovus from there*

Innovus_reports/

mult.apr_wc3_timing.txt *// top 3 worst case paths from Innovus final design.*

mult.apr_bc3_timing.txt *// top 3 best case paths from Innovus final design.*

DRC.report

LVS.report

APR_layout/

Screenshot of final layout in Innovus

Virtuoso_layout/

Screenshot of final layout in virtuoso

Verify/

Screenshot of waveforms after running the testbench on synthesized gate netlist

Screenshot of waveforms after running the testbench on placed and routed netlist with sdf back annotation

Make sure none of the items above are omitted.

Put all files in a folder. Compress the submission folder using the following command, and submit it via Canvas.

- tar -czvf ece5746_<NetID>_lab3.tar.gz <lab3_folder_containing_all_files>